

AI SOLUTION – DOCUMENTATION ASPECT

1. AI Solution

PREDICT-CARE: Predictive AI-Driven Patient Management & Clinical Care System

PREDICT-CARE is an Artificial Intelligence (AI)-driven patient management and clinical care support system designed for the healthcare industry. The proposed solution will use Python and machine learning techniques to analyse healthcare data and generate predictions and insights that can assist healthcare facilities in managing patients and their available resources more effectively.

The system will analyse relevant patient and operational data to identify patterns in areas such as patient demand, waiting times and patient management. These insights can assist healthcare staff with patient prioritisation, patient-flow management and operational planning. PREDICT-CARE is designed as a decision-support system and will not replace healthcare professionals or independently make medical diagnoses. Instead, the system will provide information that can support healthcare professionals and managers in making informed decisions.

2. Business Background

Healthcare facilities are required to manage large numbers of patients while making effective use of limited resources such as healthcare professionals, consultation areas and available time. Patient demand can vary throughout the day and over different periods, which can contribute to congestion, long waiting times and difficulties in managing patient flow.

Traditional approaches to patient and operational management may rely heavily on manual processes and historical information. This can make it difficult for healthcare facilities to identify patterns in patient demand and anticipate periods when additional pressure on services may occur.

PREDICT-CARE is proposed as an AI-based solution to assist healthcare facilities in using their available data more effectively. By analysing historical and relevant patient-

management data, the system can identify patterns and generate predictions that may assist healthcare staff and managers with patient prioritisation, patient-flow planning and resource allocation.

The solution therefore focuses not only on the use of AI technology but also on how AI can provide practical benefits to an organisation within the healthcare industry.

3. Business Objectives

Main Business Objective

To improve patient management and operational efficiency in healthcare facilities by using AI-driven predictions to support patient prioritisation, waiting-time management, patient-flow planning and resource allocation.

Specific Business Objectives

1. Improve patient prioritisation

To analyse available patient information and provide insights that can assist healthcare staff in identifying patients who may require more urgent attention.

2. Improve waiting-time management

To analyse patterns in patient demand and waiting times in order to help healthcare facilities identify periods where delays and increased patient demand may occur.

3. Improve patient-flow management

To provide data-driven insights that can assist healthcare facilities in managing the movement of patients through healthcare services and reducing unnecessary delays.

4. Support resource allocation

To provide information about expected patient demand that can assist healthcare managers in planning the use of available staff and healthcare resources.

5. Support data-driven decision-making

To transform available healthcare data into meaningful predictions and insights that can assist healthcare staff and managers in making operational decisions.

6. Improve operational efficiency

To use AI to identify patterns in healthcare operations that may assist facilities in planning their services and making more effective use of available resources.

4. Business Success Criteria

The success of PREDICT-CARE will be assessed according to whether the proposed solution is able to provide useful and reliable information that addresses the identified healthcare-management challenges.

- The system should be able to successfully process and analyse the selected healthcare dataset.
- The machine learning model should produce predictions with an acceptable level of accuracy based on the evaluation measures selected for the project.
- The system should identify meaningful patterns in patient demand, waiting times or other selected variables within the dataset.
- The predictions generated by the system should be understandable and relevant to the intended healthcare-management problem.
- The system should provide information that could assist with patient prioritisation and patient-flow planning.
- The system should demonstrate how AI-generated insights could support better resource planning and operational decision-making.
- The proposed solution should successfully demonstrate the application of Python and appropriate AI/machine-learning techniques to a healthcare problem.
- The final prototype should provide a clear demonstration of how the proposed system could benefit a healthcare organisation.

These criteria will allow the group to evaluate not only whether the AI model functions technically, but also whether the proposed solution addresses the business problem it was designed to solve.

5. Requirements

5.1 Data Requirements

The system requires a suitable healthcare dataset containing relevant information that can be used to identify patterns and develop predictions. The data must be sufficiently relevant to the selected problem and suitable for processing using Python and machine-learning techniques.

The dataset may contain variables relevant to patient management, patient demand, waiting times, patient characteristics or other operational information selected by the group.

5.2 Functional Requirements

- Import and process the selected dataset.
- Clean and prepare the data before analysis.
- Identify relevant variables and patterns within the data.
- Apply an appropriate machine-learning algorithm.
- Generate predictions based on the trained model.
- Evaluate the performance of the model using appropriate evaluation measures.
- Present the results in a form that can be understood by the intended users.
- Provide useful insights that relate directly to patient management and healthcare operations.